

WASP - Warden Autonomous Safety Platform MVP for UTM FAI Showcase 2026

Team

- Uwais — Backend Integration / Decision Engine / Dashboard
- Paen — IoT / Hardware / Sensors
- Born — Computer Vision / YOLOv8 Model

Quick Start

1. Install Python deps: `pip install -r requirements.txt`
2. Generate TTS audio: `python generate_tts.py` (needs internet)
3. Flash ESP32: Open `esp32_sensors.ino` in Arduino IDE, upload to DevKit
4. Configure XIAO: Set AP mode (see guide), upload `CameraWebServer` sketch
5. Update `wasp_backend.py` CONFIG section (COM port, phone number, API key)
6. Run: `python wasp_backend.py`
7. Open <http://localhost:5000> in browser
8. Connect laptop to “WASP-CAM-01” WiFi

File Structure

`wasp_backend.py` — Main integration (Flask + CV + Decision Engine + TTS + WhatsApp + DB)
`esp32_sensors.ino` — ESP32 firmware (DHT11 + PIR + all sensors) `generate_tts.py` — Pre-generates BM warning audio `test_cv.py` — Standalone CV test (Born) `test_serial.py` — Standalone serial test (Paen/Uwais) `requirements.txt` — Python dependencies

Showcase Day Checklist

- Laptop charged + power bank
- XIAO powered + WiFi AP working
- ESP32 wired + USB connected
- Speakers connected (loud enough for venue)
- Supervisor phone ready with WhatsApp
- Backup demo video recorded
- Dashboard tested on projector
- Pitch script practiced (3 minutes)